

Abhishek Patidar

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PROFILE

AI/ML Engineer Specialist focused on AI-driven solutions. Post-graduate student at IIIT Lucknow specializing in Data Science with expertise in ML, DL, CV, NLP, and Generative AI. Experienced in developing real-time computer vision applications for monitoring and healthcare. Passionate about leveraging AI to solve real-world challenges.

EDUCATION

- **Indian Institute of Information Technology (IIIT Lucknow)** Lucknow
Masters of Science in Data Science ; CGPA: 8.85 Aug. 2023 – May 2025
- **Shri Vaishnav Vidyapeeth Vishwavidyalaya, (SVVV Indore)** Indore
Bachelor of Science in Mathematics (Hons); CGPA: 8.46 Aug 2020 – May 2023

EXPERIENCE

AI Engineer Full time, Dermatics India, Surat June 2025 – Present

Led the development of an **AI-powered skincare recommendation system** using LLMs including Gemini AI, OpenAI GPT, and a custom-trained SkinGPT-4 for facial skin analysis and skin condition detection. Developed a multi-step pipeline incorporating product history, facial photo analysis, skincare goals, report generation, and chatbot interaction.

Implemented multi-LLM support with robust API failover handling, improving uptime. Enhanced user experience via camera-based face capture, interactive chat UI, PDF export, and real-time e-commerce product recommendations. [Website link](#)

Data Scientist Intern, Cornerstone Solutions, Indore Dec 2024–May 2025

Developed real-time computer vision solutions using YOLO models (v8, v11) and drone footage for construction site monitoring and retail, including worker/machinery detection, tracking, re-identification, and progress analysis. Key projects include:

- **Real-Time Multi-Camera Person Tracking and Re-Identification:** Designed a system to track individuals across multiple videos from different camera angles. Used YOLO for person detection, integrated with Deep SORT and ByteTrack for robust multi-object tracking. Applied Torchreid for accurate person re-identification across video sources using unique identity assignment. [Demo Video](#)
- **Theft Detection in Retail Shops:** Developed a real-time theft detection system using YOLOv11n, integrated with alert and alarm mechanisms. [Demo Video 1](#) [Demo Video 2](#)
- **Real-Time Speed Monitoring by BIDS:** Developed a system to monitor vehicle speeds in real-time using BIDS framework integrated with YOLOv8 for detection and tracking.
- **Plant Health Monitoring:** Created a drone-based CV system to detect stressed vegetation near road divergences. [Demo Video](#)
- **Real-Time Bhopal Road Monitoring System:** Built a road condition monitoring system using YOLOv11 to detect and classify various defects such as potholes, cracks, edge damage, and faded lane markings. The system helped in visual analysis and prioritization of road maintenance operations.

Data Scientist-1 Intern, MacroCosmos Creations (MCC) Private Limited, Bangalore

Oct 2024–Dec 2024

Conducted data analysis and developed a machine learning-based system to predict crop-specific irrigation needs by analyzing soil moisture, crop type, growth stage, and field conditions. Leveraged advanced ML

models to optimize water usage and support sustainable agricultural practices through data-driven decision-making.

PROJECTS

AI Agent Mental Health Therapist Assistant Project: [Demo Video](#) GitHub: [Project Link](#)

- **Objective:** Developed an AI-driven mental health assistant with **agent-based workflows**, providing safe and accessible therapy-like interaction.
- **Approach:** Built using **Google MedGemma LLM**, Ollama, and multi-API support (GPT-4, Gemini flash, Claude, Groq). Implemented an **AI Agent architecture** with **LangChain** and **LangGraph** for agentic reasoning. Designed a full-stack pipeline with **Streamlit (frontend)** and **FastAPI (backend)**. Integrated AI Agent tools including Twilio for emergency calls, location services, and professional support routing.
- **Impact:** Delivered an intelligent AI agent capable of contextual Q&A, emergency handling, and professional escalation, improving mental health support accessibility and reliability.

Cardiovascular Health Prediction Using ML GitHub: [Project Link](#) Live : [Web Application](#)

- **Objective:** Develop a framework to detect early heart disease from diverse health datasets.
- **Approach:** Data collection, EDA, feature engineering, and models such as Logistic Regression, Decision Tree, and KNN, achieving Recall and F1 Scores of 90

NLP-Powered Text Synthesis & Summarization (Model Training & Fine-Tuning)

- **Objective:** Develop an NLP framework for automatic text synthesis and summarization of lengthy conversations and documents.
- **Approach:** Implemented sequence models (RNN with LSTM) for generating concise summaries from large datasets, and fine-tuned the PEGASUS model on the Samsum dataset for conversation summarization.

GitHub: [Project Link](#)

PEGASUS Summarize: [Project Link](#)

Real World Medi-Consult Chatbot using Generative AI

GitHub: [Project Link](#)

Project: [Demo Video](#)

- **Objective:** Develop an advanced medical chatbot to provide accurate responses to health-related queries using the LLM.
- **Approach:** Utilized LangChain, Llama 2, and Pinecone for efficient querying.
- **Real-world Application:** Used in healthcare to assist patients with medical queries, provide preliminary diagnoses, and recommend treatments or further consultation, enhancing accessibility to medical knowledge.

TECHNICAL SKILLS

Languages: Python, C, LaTeX; **Data Science/ML/DL/NLP:** EDA, feature engineering, data visualization (Matplotlib, Seaborn, Power BI); supervised/unsupervised learning; neural networks (CNNs, GANs, RNNs, LSTMs, transformers); **Cloud:** AWS, Docker; **Big Data:** Hadoop, Spark; **Developer Tools:** Git, Jupyter; **Databases:** MySQL, MongoDB. **Frameworks/Tools:** TensorFlow, Scikit-learn, Keras.

ACHIEVEMENTS AND CERTIFICATIONS

- Qualified in the **Joint Admission Test for Masters (JAM)** (2023).
- **Full Stack Data Science Masters** certification by iNeuron.ai (July 2023 – July 2024). [Certificate Link](#)
- Recipient of the **Tata Capital Pankh Scholarship** (2020-2023) and **HDFC Bank Parivartan's ECSS Scholarship** (2023-2025) for academic excellence. [Certificate Link](#)